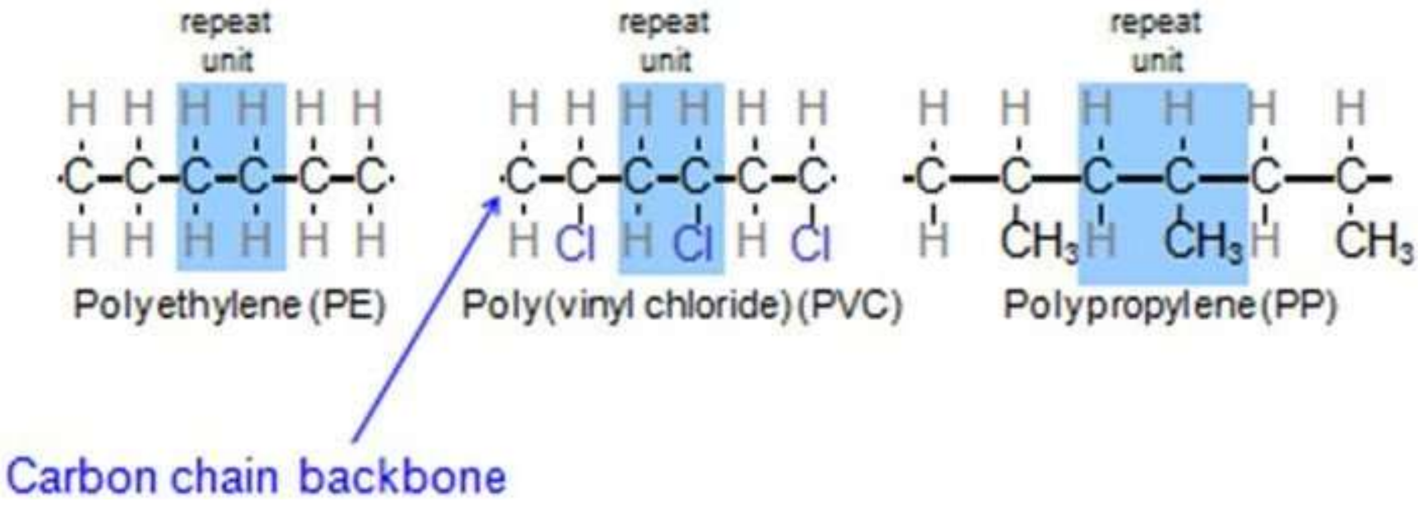
UNIT III: Engineering Materials

A) Speciality Polymers	B) Nanomaterials
Introduction, Preparation, Properties and Applications of the following polymers – 1. Engineering Thermoplastic : Polycarbonate 2. Bio-degradable polymer: Poly(hydroxybutyrate-hydroxyvalerate) 3. Conducting polymer : Polyacetylene	1. Introduction, 2.Classification of nanomaterials based on dimensions (zero dimensional, one-dimensional, two-dimensional and three-dimensional), 3. Structure, 4. Properties and 5.Applications of graphene and carbon nanotubes, quantum dots (semiconductor nanoparticles)

Polymer –

- Poly – many
- Mer – small simple, repeating unit, building block



- **Polymer:** A compound form by joining many small simple chemical units to form a long chain compound, is known as polymer.

$$n\,M \rightarrow -M-M-M-M-M-M-M-M-$$

Monomer **Polymer**

Classification of Polymer based on their occurrence

1) **Natural Polymer** - Wood, Cotton, Wool, Leather, Silk, cellulose, protein, DNA , RNA, Sugar, starch,
2) **Synthetic Polymers** - PP, PS, PE, PVC, PMMA, PF, UF, Nylon, Polyester, Teflon, Cellophane

e.g. **Nylon** first synthetic polymer

3) **Semi synthetic Polymer –** are those which are prepared from both natural and Synthetic polymer. E.g. **Rayon** is prepared from the wood and viscose, first semi synthetic polymer

- Polymers are giant molecule known as Macromolecules and have high molecular weights.
- Precursor of synthetic polymer – Bakelite – discovered by Leo Baekeland
- Bakelite –used for making telephone and electric components like electric switches
- Synthetic polymer find uses as the substitute material for – metals, woods, cotton, wool, gums, Glass, ceramics
- Polymers are use in Automobile Engg., Mechanical Engg., Electrical Engg., Civil Engg., Medical and Agriculture field

- Monomer
- Monomer is the simple chemical substance of low molecular weight which can be converted into a polymer and the molecule of monomer has at least two easily reacting positions.
- **Polymerization**
- The chemical process, in which a large number of monomer molecules get joined to form the polymer molecules, is known as Polymerization.

e.g

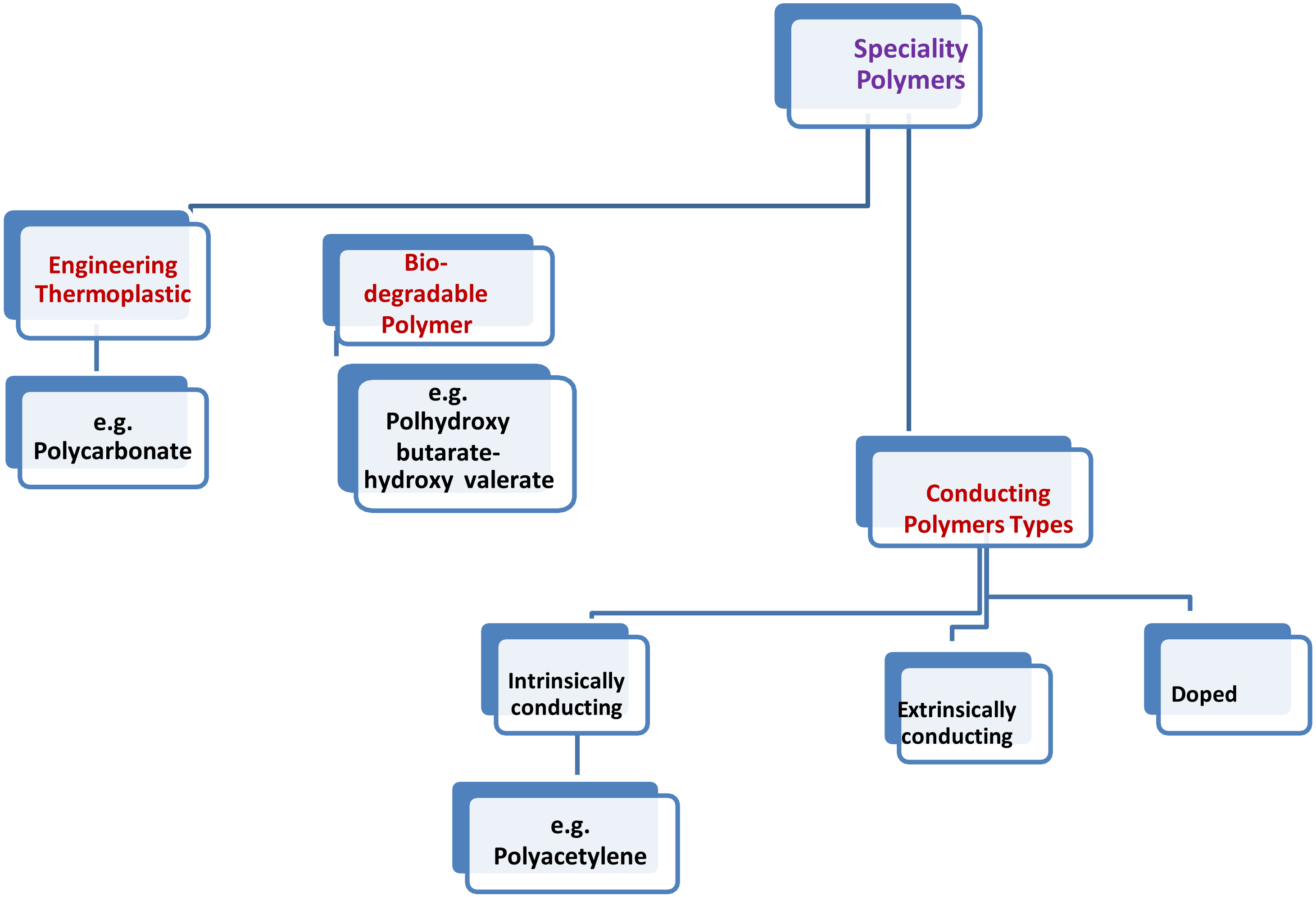
$$\underset{\text{Ethylene}}{n\,(CH_2{=}CH_2)} \xrightarrow{\text{Polymerization}} \underset{\text{Polyethylene}}{-\!\left(CH_2{-}CH_2\right)_n\!-}$$

• Classification of Polymer

1) Based on Origin - Natural, Synthetic and Semisynthetic Polymer
2) Based on atoms in the main chain of the polymer - Organic, Inorganic Polymer
3) Based on structure - Linear , branch and crosslinked polymer
4) Based on Heat treatment - Thermoplastic and Thermosetting polymers
5) Based on Polymerization reactions Addition and Condensation polymer
6) Based on number of different monomers in the polymer chain – Homopolymer and copolymer
7) Alternate copolymer, block copolymer, random copolymer,

Difference Between Thermoplastic and Thermosetting Plastic

Thermoplastic	Thermosetting Plastic
Thermoplastics are usually formed by the addition of polymerization.	Thermosetting plastics are often formed by the condensation polymerization.
It contains long-chain linear polymers and held together by weak Van der Waal forces.	It contains a 3D network structure constructed with strong covalent bonds.
Usually becomes soften on heating and stiffen on cooling.	It does not become soft on heating.
They are expensive.	They are cheap.
Thermoplastic is soluble in organic solvents.	Thermosetting plastics are insoluble in organic solvents.
They are usually soft, weak and less brittle in nature.	They are usually hard, strong and more brittle in nature.
Can be remolded.	They can't be remolded.
An example of thermoplastic is polythene.	An example of Thermosetting plastic is Bakelite.


Speciality Polymers
Engineering Thermoplastic
e.g. Polycarbonate
Bio-degradable Polymer
e.g. Polhydroxy butarate-hydroxy valerate
Conducting Polymers Types
Intrinsically conducting
e.g. Polyacetylene
Extrinsically conducting
Doped

(

1) Engineering Thermoplastics: are the group of materials obtained from high polymer resin which provide one or more outstanding properties when compared with the commodity thermoplastics such as polystyrene, polyethylene, polypropylene etc.

- Advantages of engineering thermoplastics

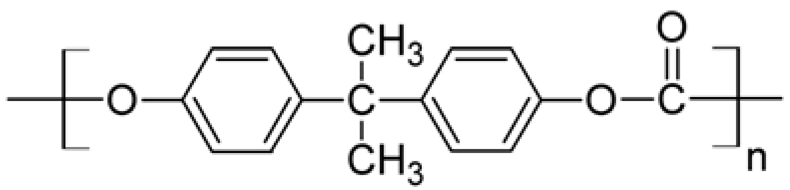
1) High thermal stability
2) Excellent chemical resistance
3) High Tensile strength
4) High impact strength
5) High flexibility
6) High mechanical strength
7) Light weight
8) Readily mouldable into complicated shapes
9) High dimensional stability

Due to these properties they can be use in place of metals, alloy, glass, ceramics- finds application in automobiles, electrical, electronics, telecommunications, textile, satellite, robots, computer components,

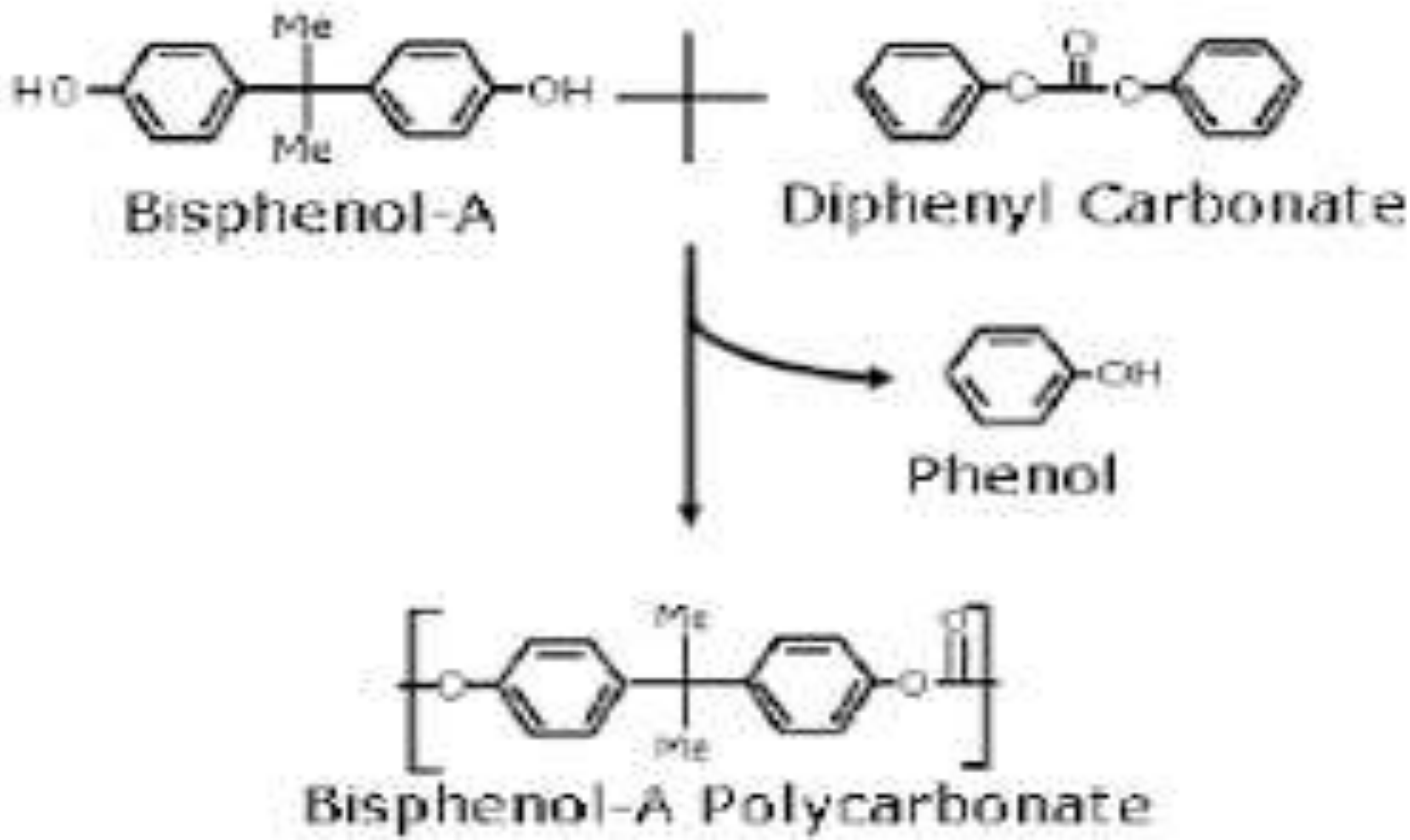
Engineering plastics are 1-5 times costly than the commodity plastics.

Engineering Thermoplastic- Polycarbonate (PC) (Lexan, Merlon)

- Polycarbonates are thermoplastic polyesters.
- Commercial names of PC are **Lexan, Merlon**
- **Polycarbonates** (PC) are a group of thermoplastic polymers containing carbonate groups in their chemical structures.
- **Polycarbonates** used in engineering are strong, tough materials, and some grades are optically transparent. They are easily worked, molded, and thermoformed.
- The most common type of polycarbonate is obtained by interacting Bisphenol A with diphyenyl carbonate.



Polycarbonate


Me
HO
OH
Me
Bisphenol-A
O
O
O
Diphenyl Carbonate
OH
Phenol
Me
O
O
O
Me
Bisphenol-A Polycarbonate

Polycarbonate

Properties- It is an amorphous polymer with attractive engineering properties such as,

1) High impact strength
2) High melting point(265ºC)
3) Very high tensile strength
4) Highly Transparent plastic, higher light transmittance (about 88%)
5) Show resistance to water and many organic solvents, but dissolves in number of organic solvents and alkali
6) Good heat resistance – useful up to 140°C
7) No resistance to UV

Polycarbonate

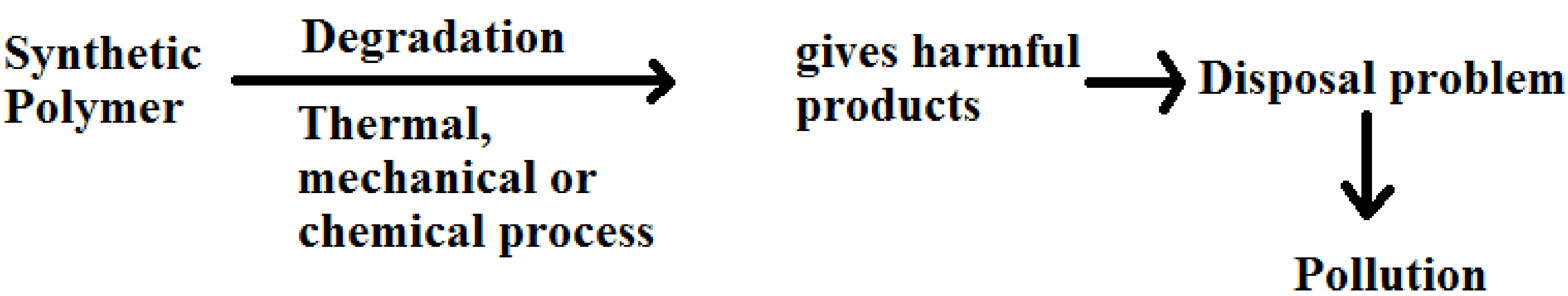
Applications –

1) **Security components-** Bullet proof material vests, for windows of vehicles and houses, Helmets, covers of vehicle lights, screw driver handles
2) **Electrical and Electronic Components-** PC being a Insulator and having heat resistance and flame retardant properties used in electrical and electronic hardware- used for making insulators, plugs, sockets, switches, covers of cell phones,
3) **Data Storage-** major application of PC is the production of CDs, DVDs and Blu-ray disc
4) **Optical applications-** due high optical clarity/ high light transmittance, used for making sunglasses, swimming and SCUBA goggles, safety goggle, visors in helmets, automotive head lamp lenses. Electronic display screens for mobile, LCD.
5) **Construction material-** due to high light transmittance, high impact and tensile strength are used as a substitute for window glass. Also used in dome lights, flat and curve glazing and sound walls.
6) **Security components –** used for bullet proof jackets, glass.
7) **Other applications –** used for housing apparatus - hair drier bodies, camera and binocular bodies.

   Moulded domestic ware such as transparent food containers, cooking utensil covers, blender jars, drinking water bottles, glasses, toys etc

2) Biodegradable Polymer

- Natural polymers- biodegradable
- Synthetic polymers and plastics- non biodegradable
- PP, PE, PC, PA – shows high resistance to microbial attack.
- **Biodegradation** of polymer is a process carried out by biological systems usually bacteria or fungi where as a polymer chain is cleaved via enzymatic activity.



- **Classification of Biodegradable polymers**

1) Natural biopolymer-

   E.g. cellulose, starch,protein

2) Biosynthetic biopolymer-

   E.g. polyhydroxy alkanotes (PHB , PHV)

3) Synthetic biopolymer – Biodegradable properties are produced synthetically

   e.g. polycaprolactone, polylactic acid.

Biodegradable Polymers

- Process by which organic substances are broken down by the environmental effects and by the living organisms.
- Biodegradable polymers are a kind of materials which degrades biologically.

Key factors for biodegradation-

a) Microorganism - Bacillus, protozoa, azetobacter, pseudomonas

b) Environment- Temp. moist condition, salt, suitable pH.

c) Nature of polymer- Hydrophilic, contains C-C single bond easily degrade rather than aromatic chains.

Polymers containing amines or carboxyllic functional groups undergo biodegradation easily as they have tendency to absorb water , swell and degrade. However, if polymer is water soluble that does not necessarily indicate that it is biodegradable.

- **Limitations of biodegradable polymers**

1. cannot manufacture on large scale
2. very costly
3. do not possess required high mechanical strength

- Features of biodegradable polymer-

1) natural polymer – biodegradable
2) synthetic addition polymers containing C atom in main chain are non biodegradable.
3) polymer chains contains atoms other than carbon in backbone – may biodegradable depending on functional group.
4) Synthetic condensation polymers – generally biodegradable.
5) amorphous polymers- biodegradable
6) hydrophilic polymers degrade faster than hydrophobic.

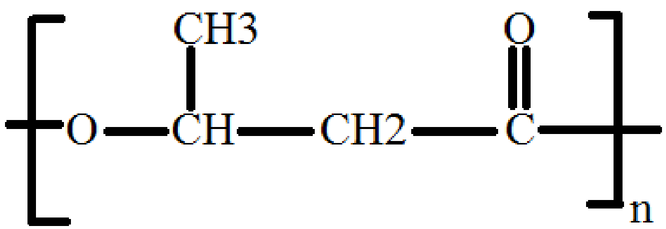
Biodegradable polymer

- Two types of polyhydroxyalkanoates (PHA) are known- Polyhydroxybutarate (PHB) and Polyhydroxyvalarate (PHV)
- PHAs are polyesters produce in nature by bacterial fermentation of sugars (sucrose, glucose and lactose) or lipids.
- Used in the production of bioplastics.
- They can be thermoplastic or elastomeric materials with melting points ranging from 40°C to 180^{o}C

Polyhydroxybutyrate (PHB)

$$\text{glucose} \xrightarrow[\text{A- eutrophus}]{\text{Fermentation}} \text{PHB}$$

- Acaligenes eutrophus (A- eutrophus)



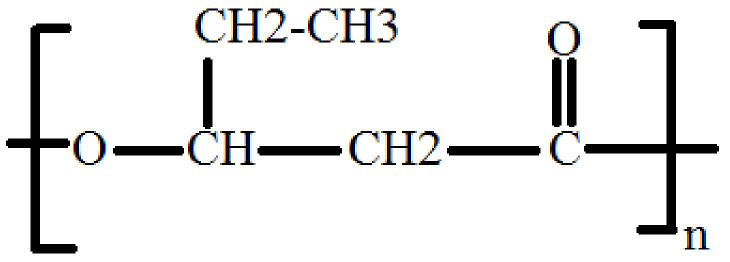
PHB

Properties

1) Brittle, highly crystalline, water soluble, non toxic,
2) Relatively resistant to hydrolytic degradation,
3) Show good O2 permeability and is rapidly biodegradable.
4) Low extension to break property
5) High M.P. (Tm = 200^{o} C)

Polyhydroxyvalarate (PHV)

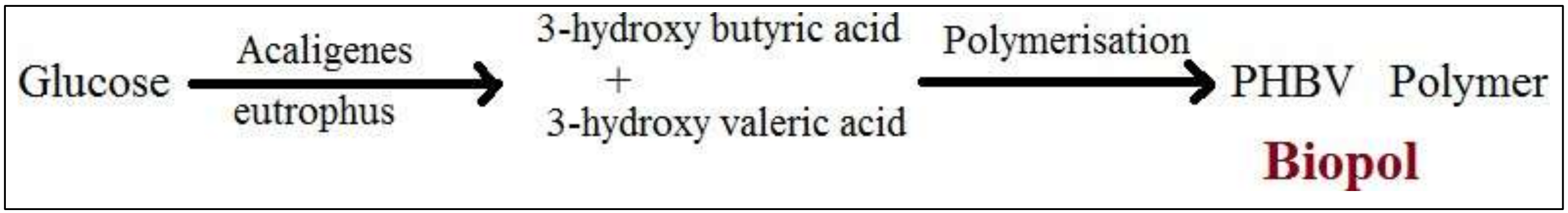
- PHV can be produced by A. eutrophous as well as another bacterium P. oleovorans depending upon the type of carbon structure available during fermentation.

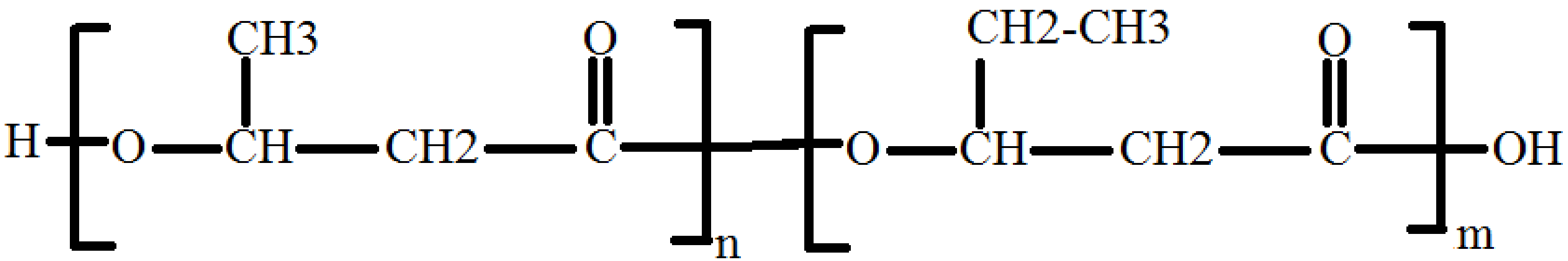


PHV

Biopol (PHBV / PHB-HV / HB-HV)

- Is a block co-polymer or random copolymer of 3-hydroxy butyric acid and 3-hydroxy valeric acid by A- eutrophus.





Biopol

Physical properties of PHBV - Biopol

- Physical properties of Bioplo copolymer vary with hydroxyvalerate content of polymer

1) as HV content increases in the range of 0.2% , polymer flexibility, toughness increases.
2) resistance of PHBV to oil is very good.
3) polymer is moisture resistant and impermeable.
4) undergo hydrolysis above pH 9 and below pH 3 it can be dissolved in chlorinated solvents like chloroform (CHCl3) and CH3Cl.

Addition of HV in HB leads to the improvement in the properties such as

1) Drop in melting temperature
2) Reduction in avearge crystallinity
3) Increased flexibility and toughness
4) Increased impact strength
5) Improve higher extension at break.

- **Biodegradation of PHBV**
- PHBV shows biodegradation in both aerobic and anaerobic condition.

$$\text{PHBV} \xrightarrow[\text{Aerobic condition}]{\text{Biodegradation}} \text{Biological material} + CO2 + H2O$$

$$\text{PHBV} \xrightarrow[\text{Anaerobic condition}]{\text{Biodegradation}} \text{Biological material} + CO2 + CH4$$

- E.g. polycaprolactone, polylactic acid, polyglycolic acid, polydioxane
- Polydioxane – biomedical application- surgical sutures
- **Applications of PHBV**

1) Medical field – controlled drug delivery system – due to PHBV biocompatibility and biodegradation propertety .

Polymer slowly degrades to smaller fragments thus release drug in controlled manner.

for internal suture, polylactic acid, polyglycolic acid, polydioxane

2) Packaging- films, blow moulded bottles, coating on papers

3) Moulded articles

4) Agriculture- **mulching** (moisture retention), **netting** (cost effective way of protecting crops from pests, birds and poultry, excessive sun rays and hailstones), **twine**, also in controlled release of fertilizer and pesticide

5) Disposable personal hygiene

Mulching
Moisture retention

Netting
(cost effective way of protecting crops from pests, birds and poultry, excessive sun rays and hailstones)

Twine

Applications of PHBV

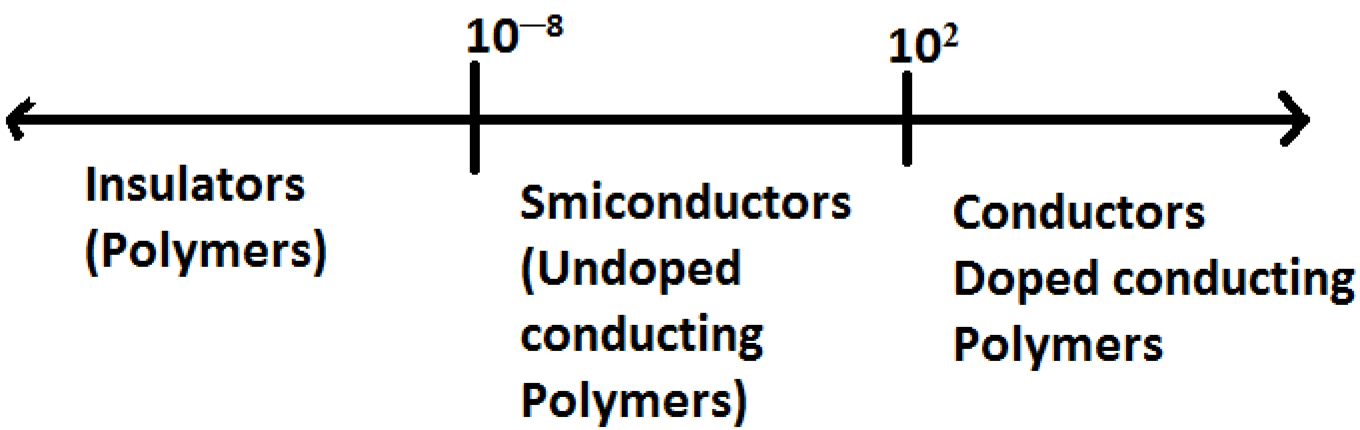
PHBV find its applications in controlled release of drugs, medical implants and repairs, specialty packaging, orthopedic devices and manufacturing bottles for consumer goods.

It has very good scope in Tissue Engineering

The favoring properties are biocompatibility and *in vivo* degradation into its monomers, which are normal in the blood. Moreover it is a bio-degradable polymer

3) Conducting Polymers

- Copper metal conductivity – 5.8 x 10^7 S/m at 20°C
- Gold metal conductivity – 4.52 x 10^7 S/m at 20°C
- In general organic polymers are bad conductor of electricity
- 1) Intrinsically conducting polymer- e.g. polyacetylene, polypyrrole, polyaniline, polyparaphenylene

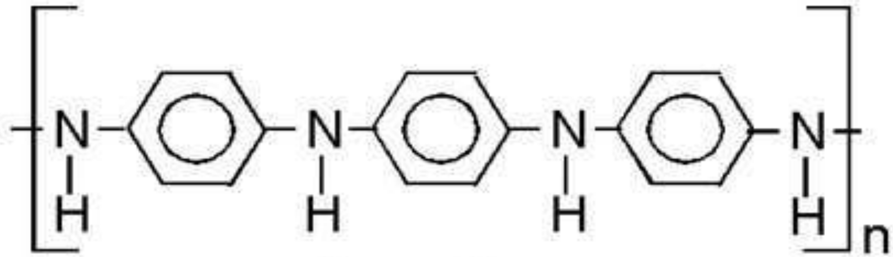


Conducting polymers

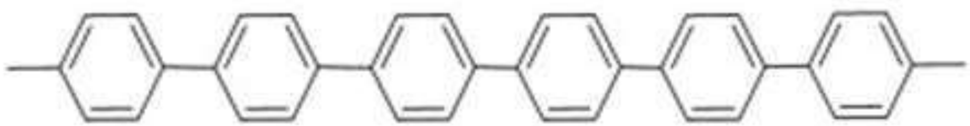
- If plastic is compounded with metal powder it becomes a conducting material but it is not by the plastic but due to polymer.
- Structural requirement

1) The Polymer chain containing conjugated (alternative) double bonds throughout its chain, which conduct electricity due to mobile electrons (delocalized pi electrons)
2) Highly crystalline polymers with and highly planar structure
3) Presence of aromatic rings in the chain with continuous resonance, increases conductivity
4) Linear chain polymer.

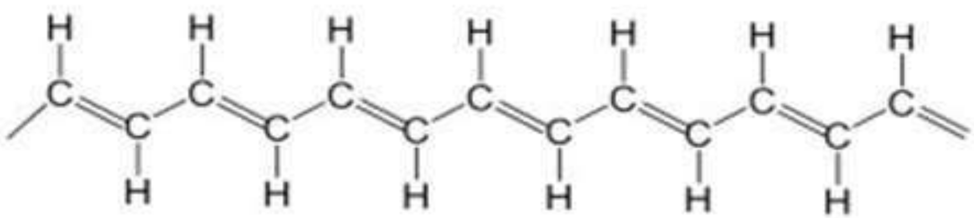
1) Examples of Intrinsic conducting polymer



Polyaniline



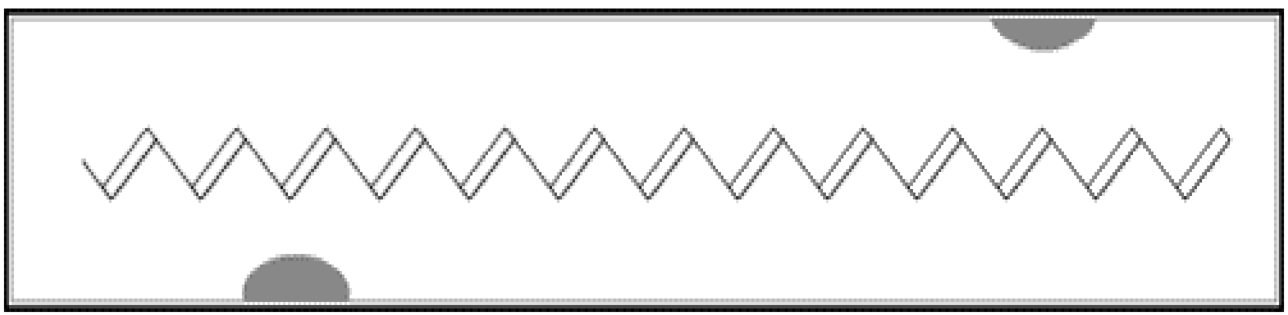
Polyphenylene



Polyacetylene

2) Extrinsic conductors

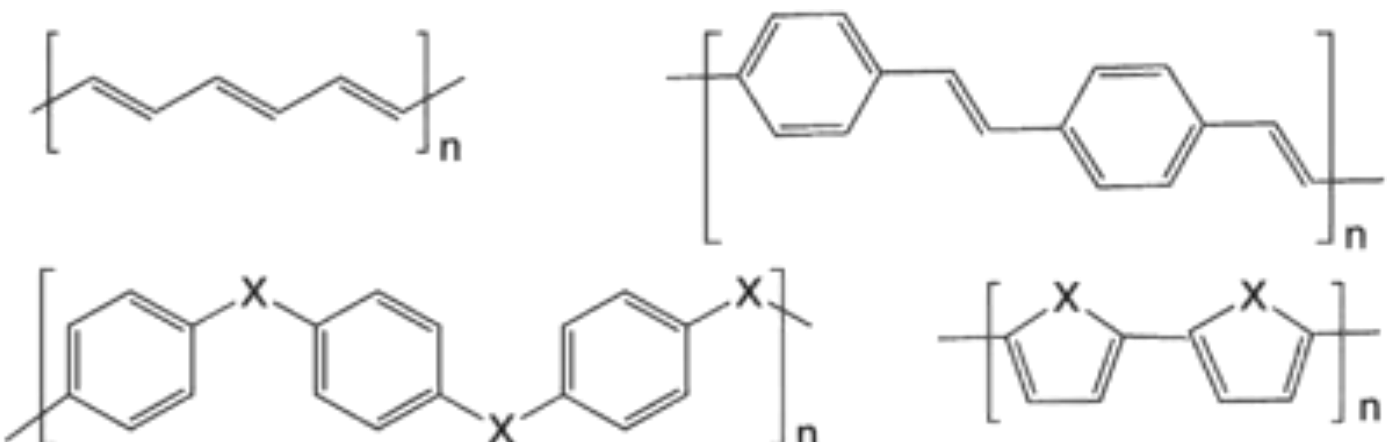
- Polymers are made conducting by Doping .
- 1) p- doping or oxidative doping – Lewis acid, I_2, Br_2, $FeCl_3$, PF_6, AsF_5 can be used as a p- dopants.
- $$-(C2H2)_n- + I_2 \xrightarrow{\text{Oxidation}} -(C2H2)_n^{+} \,.\, I_3^{-}$$
- Positive charge is developed on polymer chain.
- Polymer molecule actually looses the pi electron to halogen atom. Iodine molecule attracts electron from polymer molecule and forms I_3^- .
- The positively charge polyacetylene molecule is radical cation or ***polaron***.
- E.g. Doped Polyacetylene



2) n-doping or reductive doping –

- Lewis base, Na, Li, K
- Reducing agents donate a pair of electron to polymer chain. This make polymer chain negatively charge and make it conducting.

$$\left(\mathrm{C2H2}\right)_n + \mathrm{Na} \xrightarrow{\text{Reduction}} \left(\mathrm{C2H2}\right)_n^{-}\ \mathrm{Na}^{+}$$



(X = NH/N, S) (X = NH, S)

Applications of conducting polymers

1. Rechargeable batteries- promising charge storage material. Batteries are small size, long lasting, produce high current density
2. Optical filters- radiations from computer screen can be absorbed by conductive polymer

3. Sensors- pH, O2, NO2, SO2, NH3, glucose

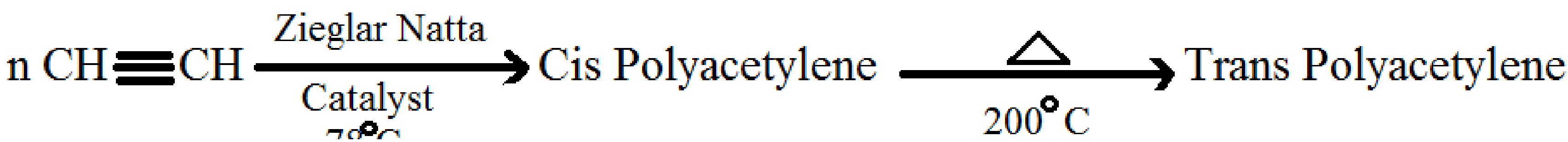
Tri-iodide oxidized polyacetylene – used as sensor to measure glucose concentration

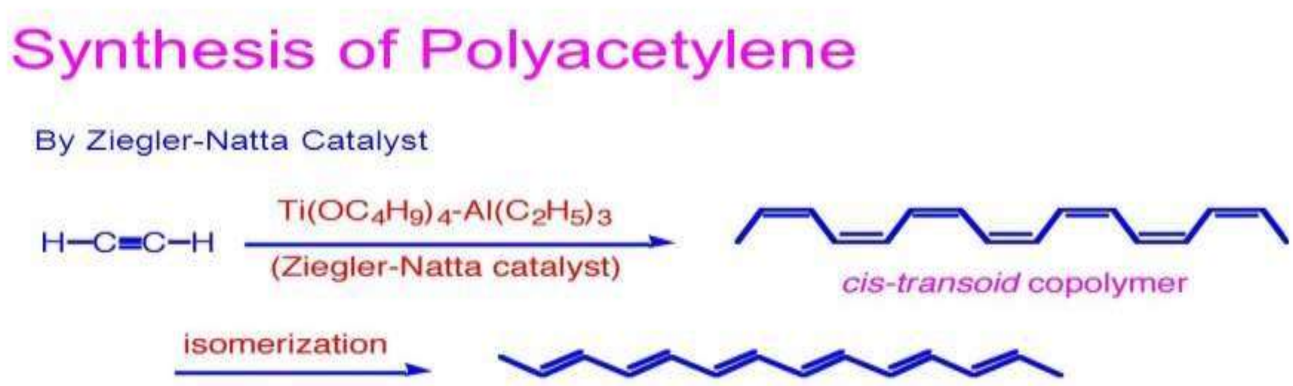
4. In electronics- LED, rattemitting diodes, data storage, light emitting wall papers.

5. photovoltaic cells

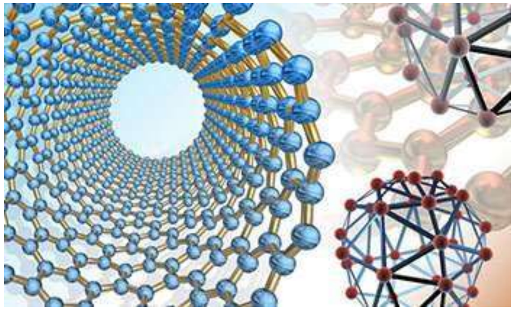
6. telecommunication systems

Polyacetylene





- Order of conductivity
- Cis < Trans < doped ethylene
- Cis polyacetylene conductivity → 1.7 x 10^{-9} S/cm
- Trans polyacetylene conductivity → 4.4 x 10^{-5} S/cm
- Doped polyacetylene → 400 S/cm
-



B) Nanomaterials

Definition of Nanomaterials –

Nanomaterials are defined as a set of substances where at least one dimension is less than approximately 100 nm.

Nanomaterials are characterized by a very small size in the range of 1-100 nm.

- A nanomaterial is one millionth of a millimetre approximately 100,000 times smaller than the diameter of human hair.
- Nano scale material have unique optical, magnetic electrical and mechanical properties. These emerging properties have great impact in electronics, medicine, energy storage and other fields.
- Nanomaterials used in engineering field are engineered nanomaterials- they are designed at the molecular (Nanometre) level . These nanomaterials shows novel properties which are generally not seen in their conventional bulk material.

- Reasons for different properties in Nanomaterials than their bulk materials
- Size of atom ranges from 0.1 to 0.5 nm i.e. $1x10^{-10}$m to $5x10^{-10}$ m
- Structural features of nanomaterials are in between those of atoms and the bulk materials.
- Different properties of nanomaterials is due to the nanometer size of the materials which is responsible for following properties.

1) It has large fraction of surface atoms
2) They have high surface energy
3) They have special confinement
4) They have reduced imperfection

- The properties of nanomaterials are different and special fro their bulk material is due the following reasons-

A) Much greater surface area to volume ratio than their conventional forms - Nanomaterials due to their very small dimension have extremely large surface area to volume ratio, resulting in more surface dependent properties. E.g. Metallic nanoparticles can be used as very active catalysts. Chemical sensors from nanoparticles and nano wires enhance the sensitivity and sensor selectivity.

B) Quantum effect- The nanometre size of nanomaterial also have spatial confinement effect on the materials which brings the quantum effects. The energy band structure and charge carrier density in the materials can be modified quite differently from their bulk and in turn will modify the electronic and optical properties of the materials. For example, lasers and Light Emitting Diodes (LED) from both of the quantum dots and quantum wires are very promising in the future optoelectronics. High density information storage using quantum dot devices is also a fast developing area.

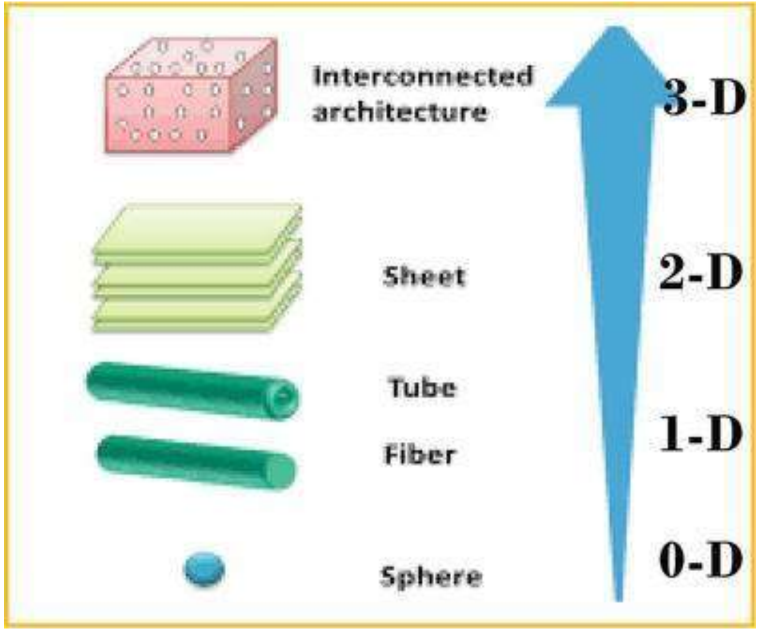
C) Remduced Iperfections:- This is also an important factor in determination of the properties of the nanomaterials. Nanostructures and nanomaterials favours a self-purification process in that the impurities and intrinsic thermal annealing which in turn increases the perfection in materials and thus affects the properties of nanomaterials . For example, the chemical stability for certain nanomaterials become better than the bulk materials.

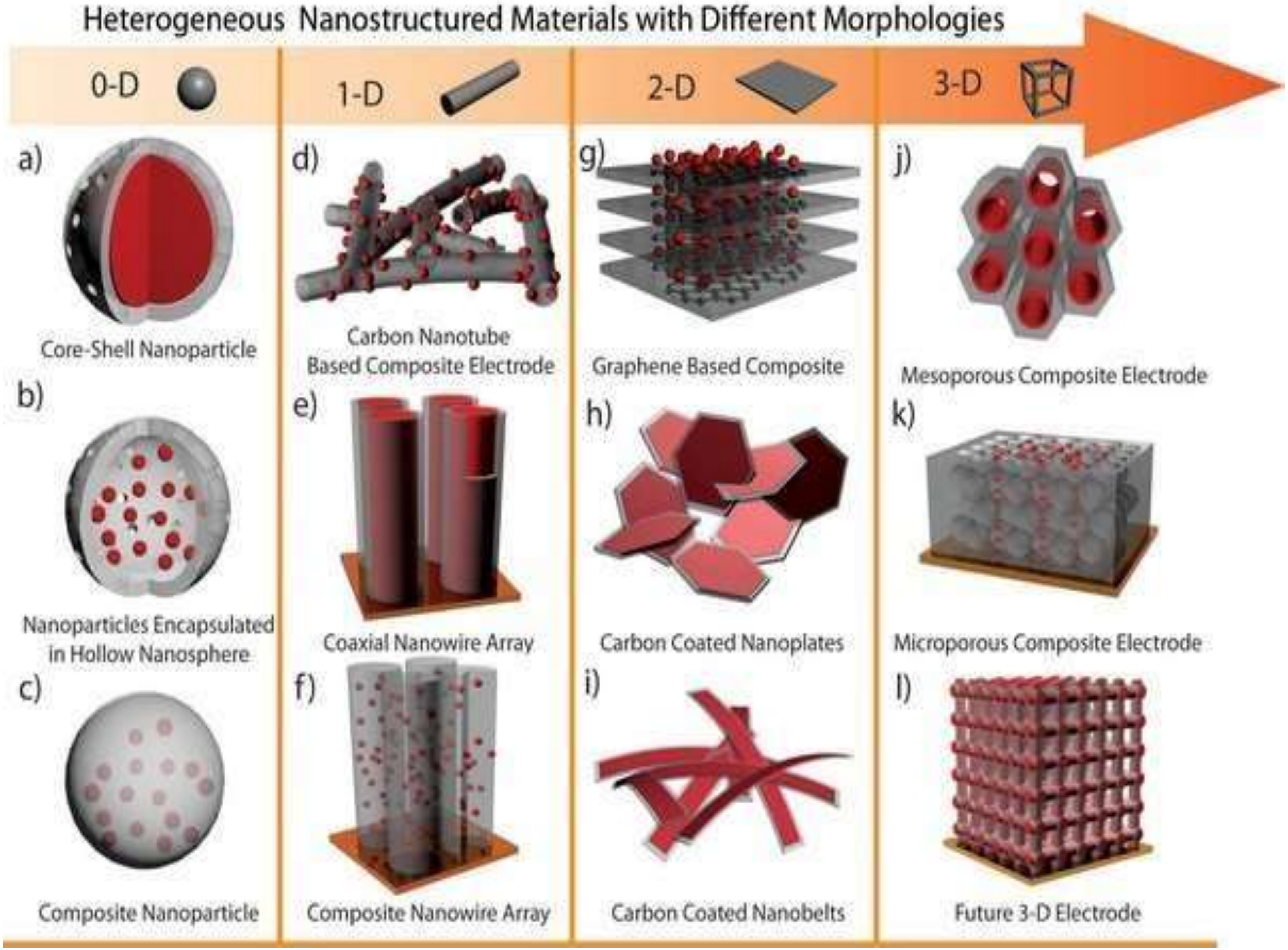
Nanomaterials

Classification of Nanomaterials

Nanomaterials can be classified as

1) **Zero dimensional (0 D) –** these are the materials having all the dimensions are measured within nanoscale i.e. no dimensions are larger than 100 nm. e.g. nanoparticles, quantum dots.
2) **One dimensional (1D) –** these are the materials having two out of three dimensions are measured within nanoscale i.e. no dimensions are larger than 100 nm. e.g. nanowires, nanorods, nanoneedles and nanotubes.
3) **Two dimensional (2D) –** these are the materials having one out of three dimensions are measured within nanoscale i.e. no dimensions are larger than 100 nm. e.g. nanocoatings, nanofilms
4) **Three dimensional (3D) –** these are the materials having all three dimensions not measured within nanoscale i.e. no dimensions are larger than 100 nm. e.g. nanocrystals, nano composite materials.





Heterogeneous Nanostructured Materials with Different Morphologies
0-D
1-D
2-D
3-D
a)
Core-Shell Nanoparticle
b)
Nanoparticles Encapsulated in Hollow Nanosphere
c)
Composite Nanoparticle
d)
Carbon Nanotube Based Composite Electrode
e)
Coaxial Nanowire Array
f)
Composite Nanowire Array
g)
Graphene Based Composite
h)
Carbon Coated Nanoplates
i)
Carbon Coated Nanobelts
j)
Mesoporous Composite Electrode
k)
Microporous Composite Electrode
l)
Future 3-D Electrode

(Properties of Nanomaterials) Nanoscale Size Effect

- Manifestation of novel phenomena and properties, including changes in:
  - Physical Properties (e.g. melting point)
  - Chemical Properties (e.g. reactivity)
  - Electrical Properties (e.g. conductivity)
  - Mechanical Properties (e.g. strength)
  - Optical Properties (e.g. light emission)
  - Magnetic Properties (e.g. Coercivity)

A. Introduction

Physical Properties of Nanomaterials

- Nano-sized condensed matter exhibits some remarkable specific properties that may be significantly different from the physical properties of bulk materials. Some are known, but there may be a lot more to be discovered.
- Origins : (i) large fraction of surface atoms,
  (ii) large surface energy,
  (iii) spatial confinement, and
  (iv) reduced imperfections.

Physical Properties of Nanomaterials

1. **Reduced Melting Point --** Nanomaterials may have a significantly lower melting point or phase transition temperature and appreciably **reduced lattice constants (spacing between atoms is reduced), due to a huge fraction of surface atoms** in the total amount of atoms.

2. **Ultra Hard --** Mechanical properties of nanomaterials may reach the theoretical strength, which are one or two orders of magnitude higher than that of single crystals in the bulk form. The enhancement in mechanical strength is simply **due to the reduced probability of defects**.

3. **Optical properties** of nanomaterials can be significantly different from bulk crystals.
   --- **Semiconductor Blue Shift** in adsorption and emission **due to an increased band gap**.
   Quantum Size Effects, Particle in a box.
   --- **Metallic Nanoparticles Color Changes** in spectra **due to Surface Plasmons Resonances**
   Lorentz Oscillator Model.

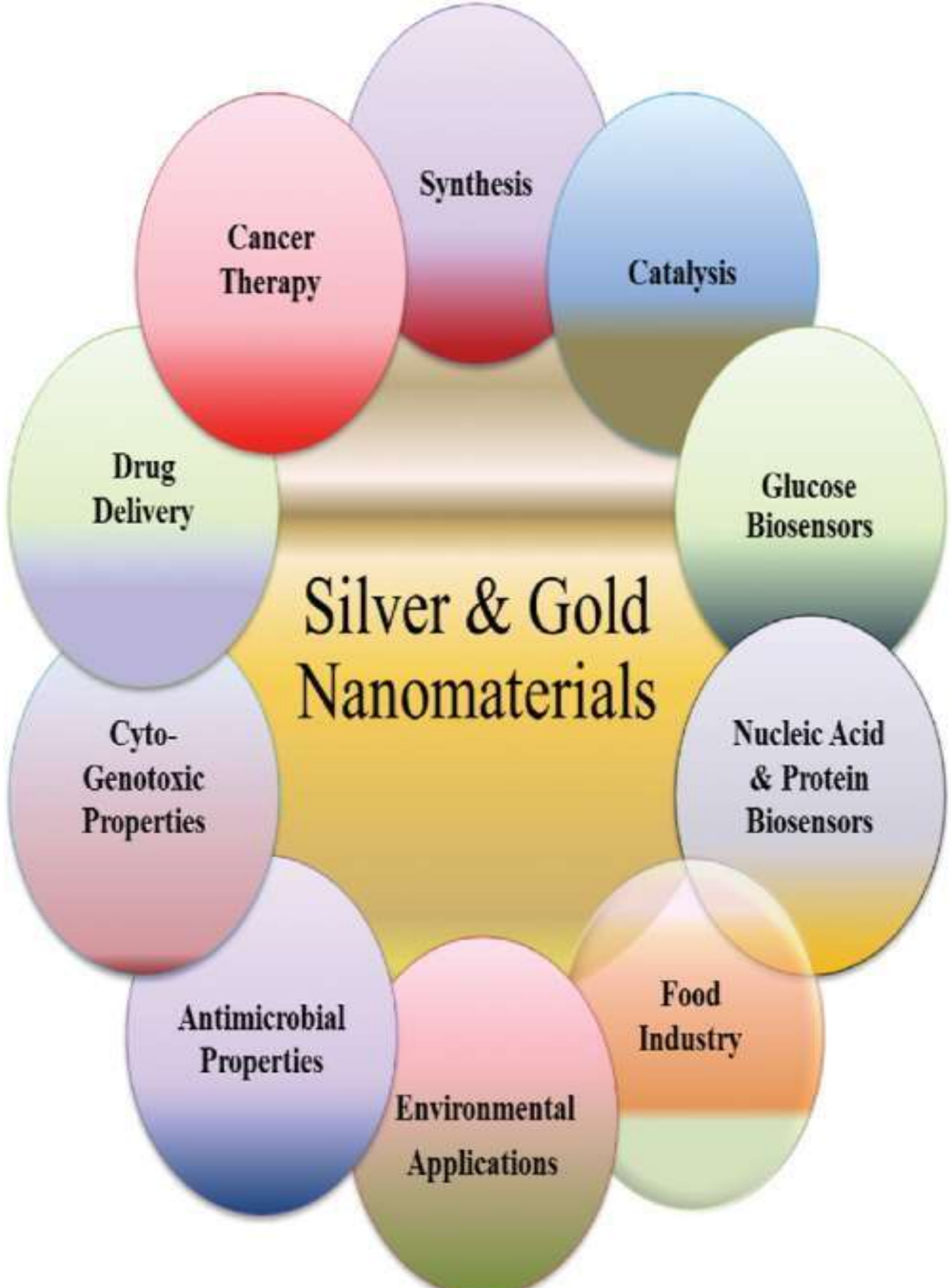
4. **Electrical conductivity decreases** with a reduced dimension **due to increased surface scattering**. **Electrical conductivity increases** due to the **better ordering** and **ballistic transport.**

5. **Magnetic properties** of nanostructured materials are distinctly different from that of bulk materials. **Ferromagnetism disappears** and transfers to **superparamagnetism** in the nanometer scale **due to the huge surface energy**.

6. **Self-purification** is an intrinsic thermodynamic property of nanostructures and nanomaterials **due to enhanced diffusion of impurities/defects/dislocations** to the nearby surface.

7. **Increased perfection enhances chemical stability.**

Most are tunable with size!


Synthesis
Cancer Therapy
Catalysis
Drug Delivery
Glucose Biosensors
Silver & Gold Nanomaterials
Cyto-Genotoxic Properties
Nucleic Acid & Protein Biosensors
Food Industry
Antimicrobial Properties
Environmental Applications

• Important Properties of Nanomaterials

a) Optical properties: - One of the most fascinating and useful property of nanomaterials is their optical properties. Various application of optical properties of nanomaterials include their use in optical detector, laser, sensor, imaging, phosphor, display, solar, cell, photocatalysis, photo electrochemistry and biomedicine.

The optical properties of nanomaterials depende on parameters such as size, shape, surface, characteristics and other variable including doping and interation with the surrounding environment or other nanostructures.

The shape can have dramatic influence on optical properties of metal nanostructures. For example, simple change in size of CdSe semiconductor to nano-size particle alters the optical properties of the CdSe nanoparticles as seen in figure.

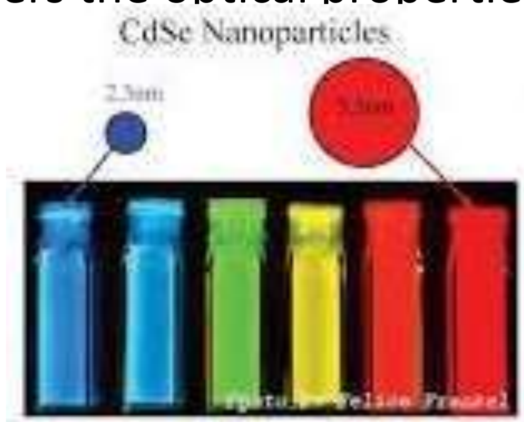


Fig. Fluorescence emission of Cadmium Selenide (CdSe) quantum dots of various sizes nanoparticles.

b) Electrical properties:- Electrical properties of nanomaterials are different then their bulk materials. In nanomaterials especially nanotubes/wires with decreasing diameter of the wire, the number of electron wave modes contributing to the electrical conductivity become increasingly smaller by well-defined quantized steps.

For example, in electrically conducting carbon nanotubes only one electron wave mode is observed which transport the electrical current.

c) Mechanical properties:- Mechanical properties of metallic and ceramic nanomaterials are influenced by porosity , given size and the filter used.

For example, filling polymers with nanoparticles or nanorods and nanotubes respectively, leads to significant improvements in their mechanical properties. Such improvements depend heavily on the type of the filter and the way in which the filling is conducted.

Compositors consisting of a polymer matrix and silicates exhibit excellent mechanical and thermal properties.

d) Magnetic properties:- Bulk gold and Pt are non-magnetic, but at in their nano size they are magnetic. Surface atoms are not only different to bulk atoms but they can also be modified by interaction with other chemical species by **capping them with nanoparticles.** This phenomenon helps in modification of the physical properties of the nanoparticles by capping them with appropriate molecules.Thus by using capping, it should be possible that non-ferromagnetic bulk materials exhibit ferromagnetic like behavior when prepared in nano range.

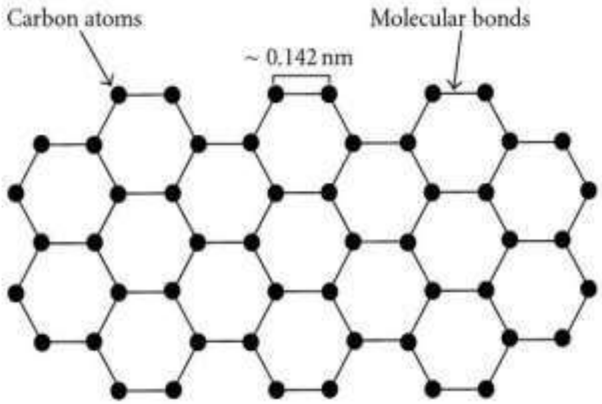
For example, magnetic nanoparticles of Pd, Pt and Au (that is diamagnetic in bulk) are obtained from non-magnetic bulk materials. In the case of Pt and Pd, the ferromagnetism arises from the structural changes associated with size effects where as in case of gold nanoparticles they become ferromagnetic when they are capped with appropriate molecules.

- **Carbon based nanomaterials**

1) **Graphene**
2) **Carbon nano tube (CNTs)**

1) Graphene –

- **Structure -** Graphene is the name for a honeycomb sheet of carbon atoms. It is a single layer of carbon atoms organized in a hexagonal lattice. It is two dimensional material. It is the building block for other graphitic materials (since a typical carbon atom has a diameter of about 0.33 nanometers, there are about 3 million layers of graphene in 1 mm of graphite)
- All carbons in graphene are sp2 hybridised
- C= $1S^2\ 2Px^1 2Py^1\ 2pz^1$
- Each Sp2 hybridized carbon is attached to three other SP2 carbon forming a hexagonal sheet and each C has one electron in unhybridized orbital which can delocalise on the sheet due to resonance, making graphene good conductor of electricity.

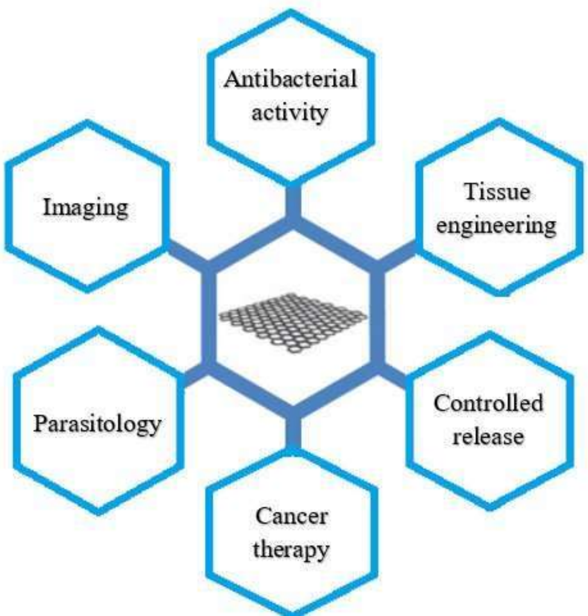


- **Properties**

1) Harder than diamond yet more elastic than rubber; tougher than steel yet lighter than aluminium.
2) Graphene is the strongest known material.
3) Graphene is the world's strongest material, and so can be used to enhance the strength of other materials.
4) Its high electron mobility is 100 times faster than silicon; it conducts heat 2x better than diamond; its electrical conductivity is 13x better than copper.

- **Applications graphene -**

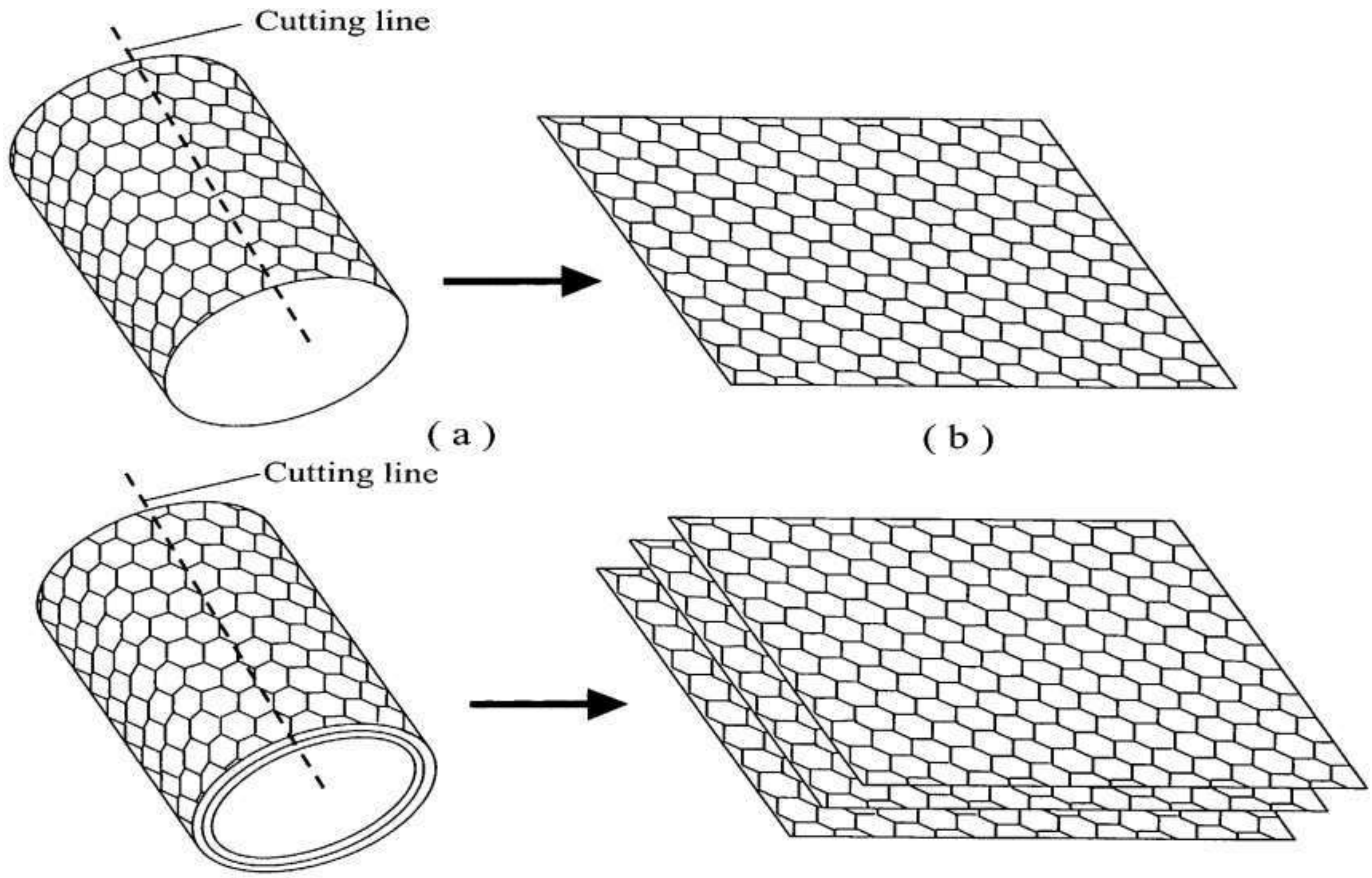
1) Graphene is use as a sensor for gas detection, also use as biosensors.
2) Graphene is used in tissue engineering. It has been used as a reinforcing agent to improve the mechanical properties of biodegradable polymeric nanocomposites for engineering bone tissue applications
3) Its powder can be used in making polymer composites.
4) Used for drug delivery agents for cancer therapy.
5) Being transparent and flexible conductor, it can be used for solar panels, LEDs, tourchpanels, smart phones.
6) Energy storage material.
7) As filtration material



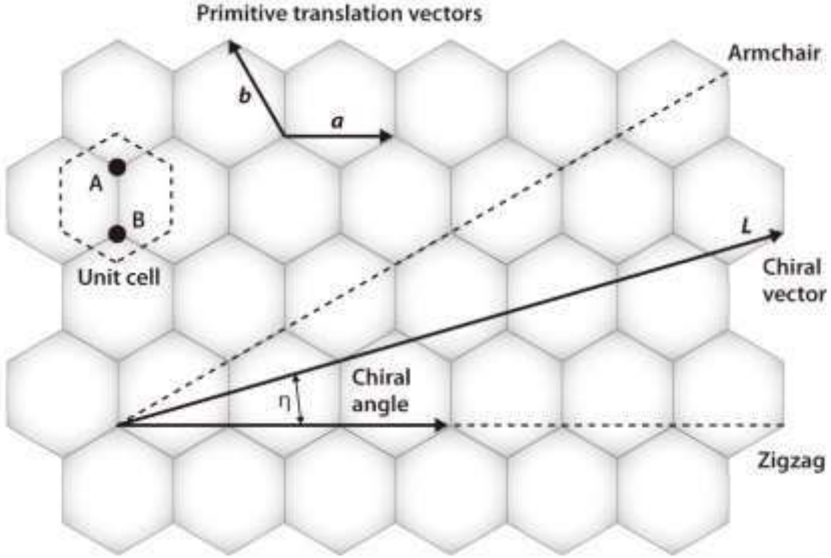
CARBON NANOTUBES

- Nano material – 1 to 100 nm dimension of material particles
- Structure: CNT s are the members of fullerene family and can be imagined as cylinders formed by rolling graphene sheet and caping by hemisphere structure similar to fullerene
- Diameter of CNT is in the order of few nm
- Armed Chair, Zigzag and Chiral are the three types of carbon nanotubes
- Nanotubes are also categorised as single wall and multiwall nanotubes
- Preparation-

1) Arc vapourisation of graphite
2) Chemical vapour deposition
3) Pyrolysis of hydrocarbon
4) Laser ablation



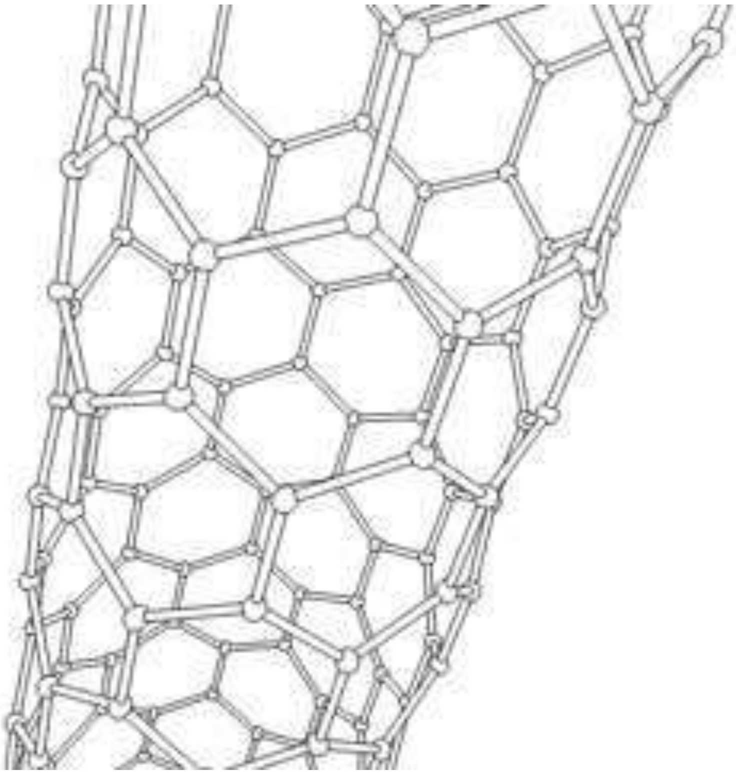
(a) (b)
(c) (d)



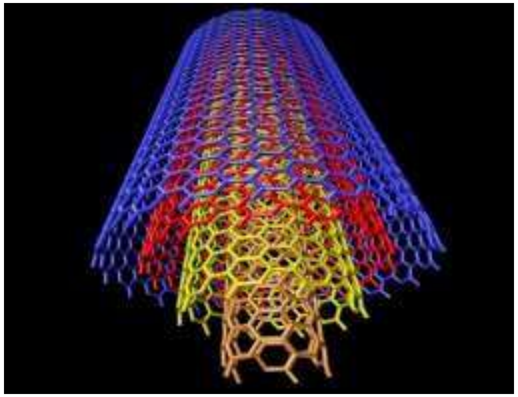
armchair

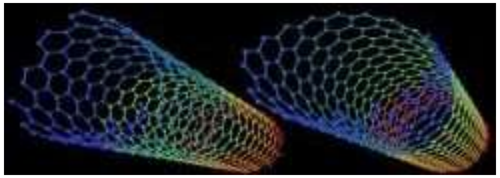
zigzag

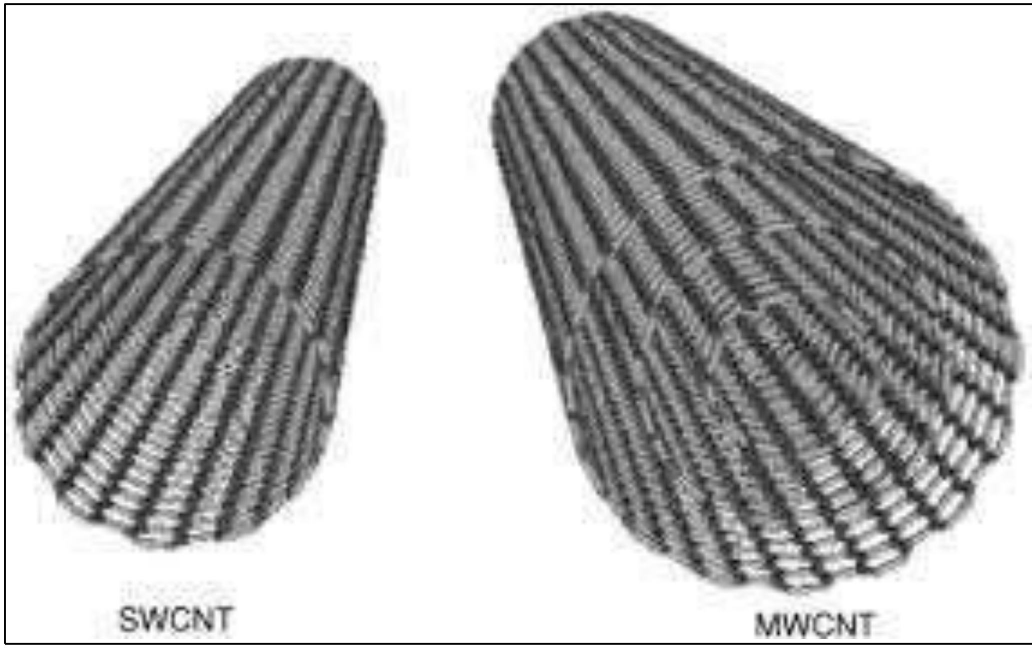
chiral



Rotating Single Walled Zigzag Carbon Nano tube



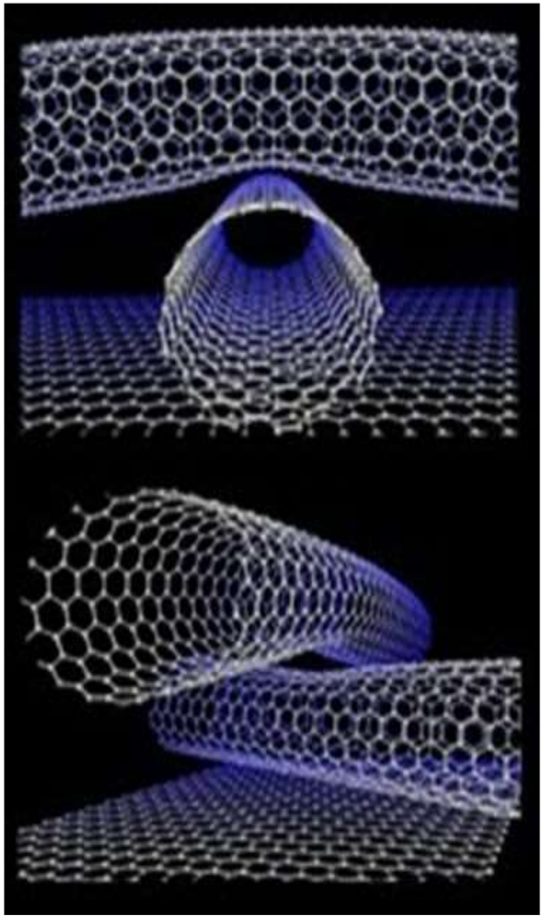




CARBON NANOTUBES

- Properties –SWCNT can be twisted flattened and bent into small circles
- The intrinsic mechanical and transport properties make them ultimate carbon fibres
- Unique combination of stiffness, strength and tenacity
- very high thermal and electrical conductivity
- Large surface area and low density

Properties of Carbon nanotubes



•the highest elastic module, and mechanical strength that is approximately 200 times stronger than steel.

•novel electronic properties.

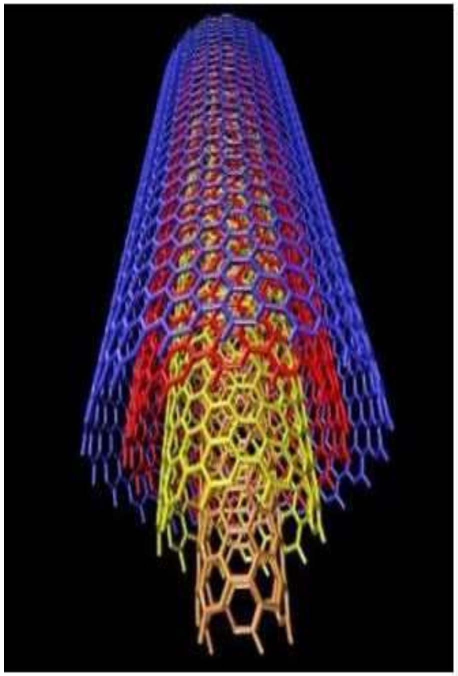
•high thermal conductivity.

•excellent chemical and thermal stability.

•promising electron field emission properties.

•high chemical (such as lithium) storage capacity.

CNT properties



- Outstanding electrical conductivity (six times copper)

- Outstanding mechanical properties (tensile strength, yet flexible)

- Field emitters (they can increase resolution in spectroscopy)

- One-dimensional thermal conductivity

- Easy functionalization

- Enhanced mass transport through the nanotube

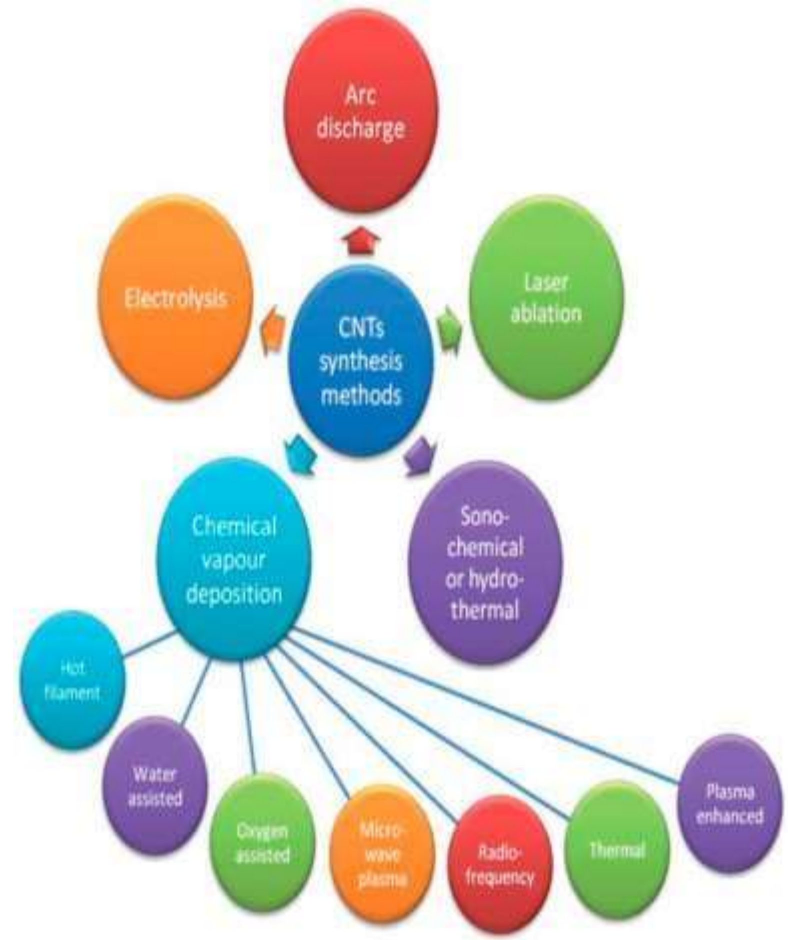
Applications of CNTs

1. Their composites can be used to build lightweight spacecraft
2. In Drug delivery system
3. In stereo -specific reaction
4. Filteration
5. In Hydrogen storage device
6. For waterproof or crease resistant fabrics
7. Stronger and lighter sports equipments like tennis, rackets, bicycle parts etc.
8. As a conductor
9. Masks
10. Body part implantation
11. Catalysts

Synthesis of CNTs

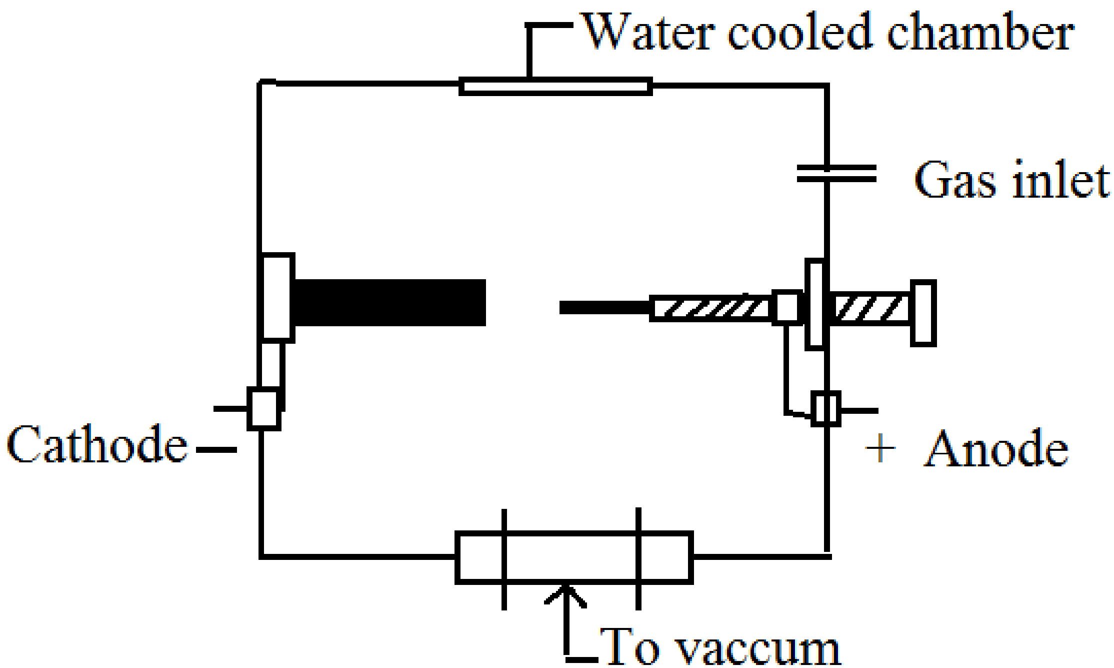
- By different methods

1) Arc vaporization of graphite
2) Chemical vapour deposition
3) Pyrolysis of hydrocarbons
4) Laser ablation or evaporation



1) Arc vaporization of graphite

- An electric arc is stuck between two graphite electrodes under following conditions
- Electrodes – graphite
- Diameter of electrode - 5 to 25 mm
- Gap between the electrode - 1mm
- Current - 50 to 100 amp
- Voltage – 15 to 25 volts
- He gas pressure - 1000 to 500 torr



- When electric arc is struck the temperature reaches to about 3000ºC
- An **electric arc**, or **arc discharge**, is an electrical breakdown of a gas that produces an ongoing plasma discharge, resulting from a current through normally nonconductive media such as air.

QUANTUM DOTS

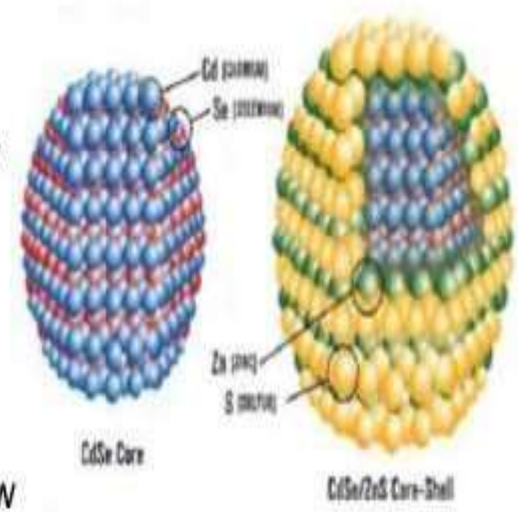
- A **quantum dot** is a nanocrystal made of semiconductor materials that are small enough to exhibit quantum mechanical properties.
- Specifically, its excitons are confined in all three <u>spatial dimensions</u>.

What is Quantumdots?

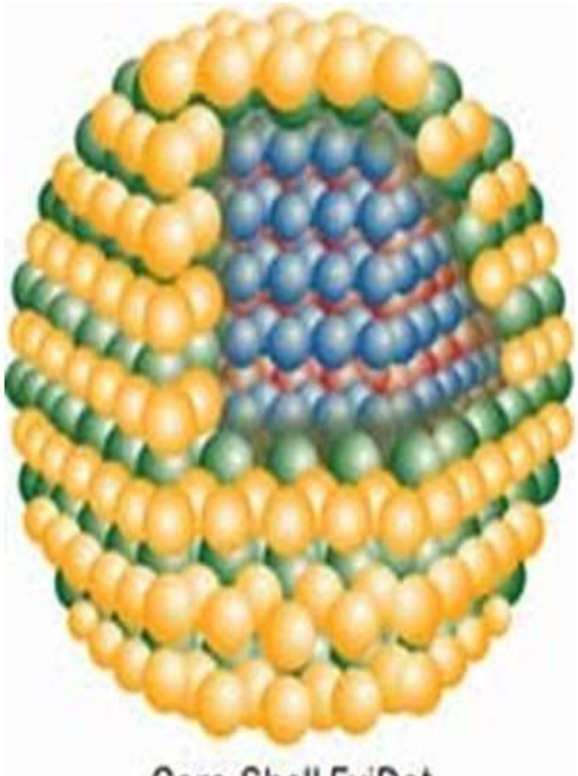
- Quantum dots are semiconductor nanocrystals.

- They are made of many of the same materials as ordinary semiconductors (mainly combinations of transition metals and/or metalloids).

- Unlike ordinary bulk semiconductors, which are generally macroscopic objects, quantum dots are extremely small, on the order of a few nanometers. They are very nearly zero-dimensional.



What is a quantum dot?



Core-Shell EviDot

- Nanocrystals
- 2-10 nm diameter
- semiconductors

Quantum Confinement: What does it mean?

- Change of electronic and optical properties when the material sampled is of sufficiently small size - typically 10 nanometers or less
- QDs are bandgap tunable by size which means optical and electrical properties can be engineered to meet specific applications
- Smaller QDs have a large bandgap
- Typically, smaller QDs (e.g., radius of 2~3 nm) emit shorter wavelengths generating colors such as violet, blue or green. While bigger QDs (e.g., radius of 5~6 nm) emit longer wavelengths generating colors like yellow, orange or red.
- Absorbance and luminescence spectrums are blue shifted with deceasing particle size

What are Quantum Dots?

- Quantum dots are semi-conductors that are on the nanometer scale.
- Obey quantum mechanical principle of quantum confinement.
- Exhibit energy band gap that determines required wavelength of radiation absorption and emission spectra.
- Requisite absorption and resultant emission wavelengths dependent on dot size.

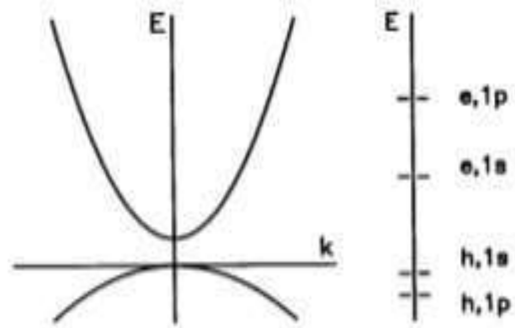
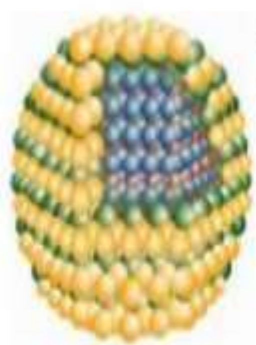
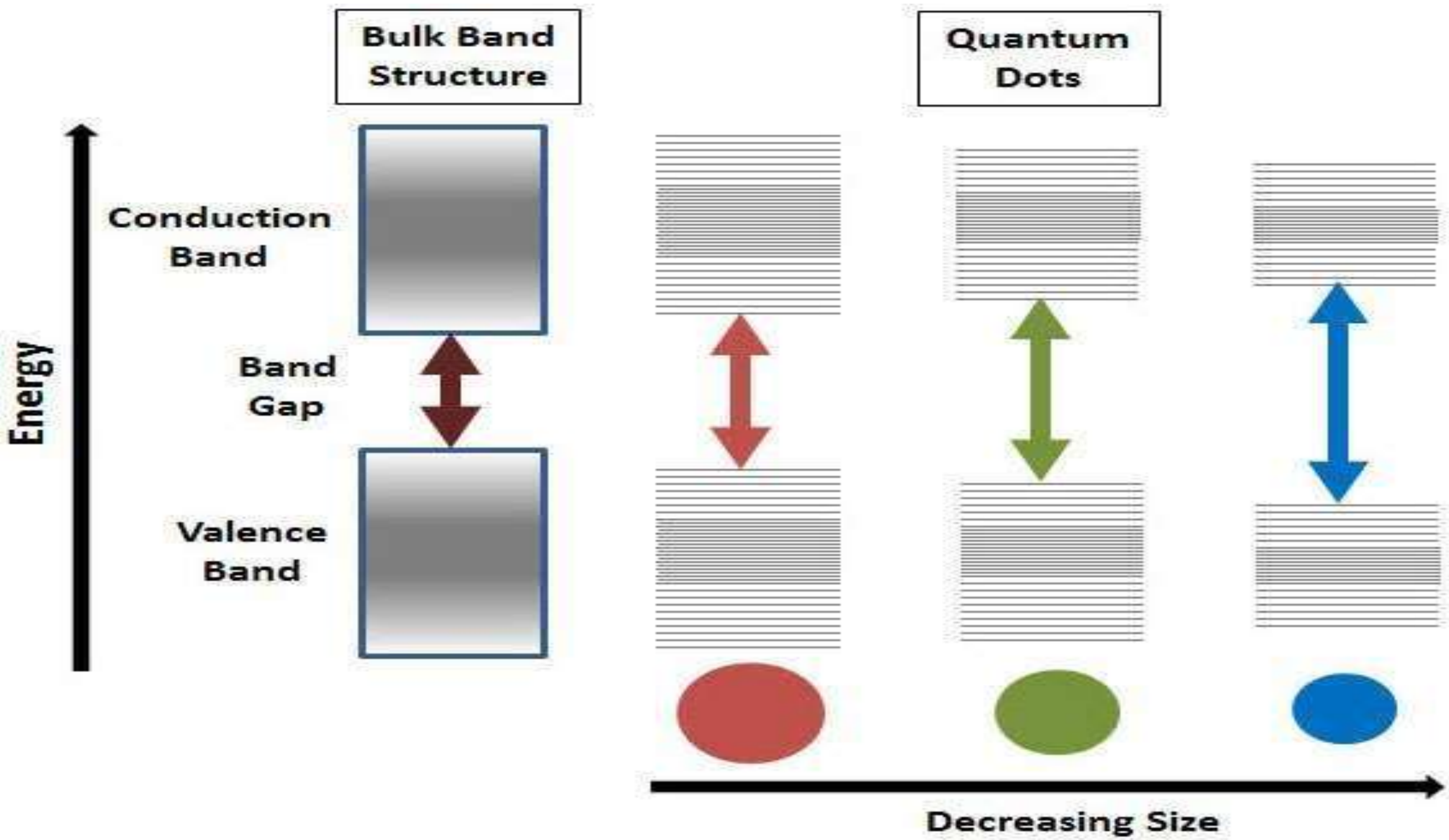


Figure: Schematic plot of the single particle energy band gap. The upper parabolic band is the conduction band, the lower the valence.



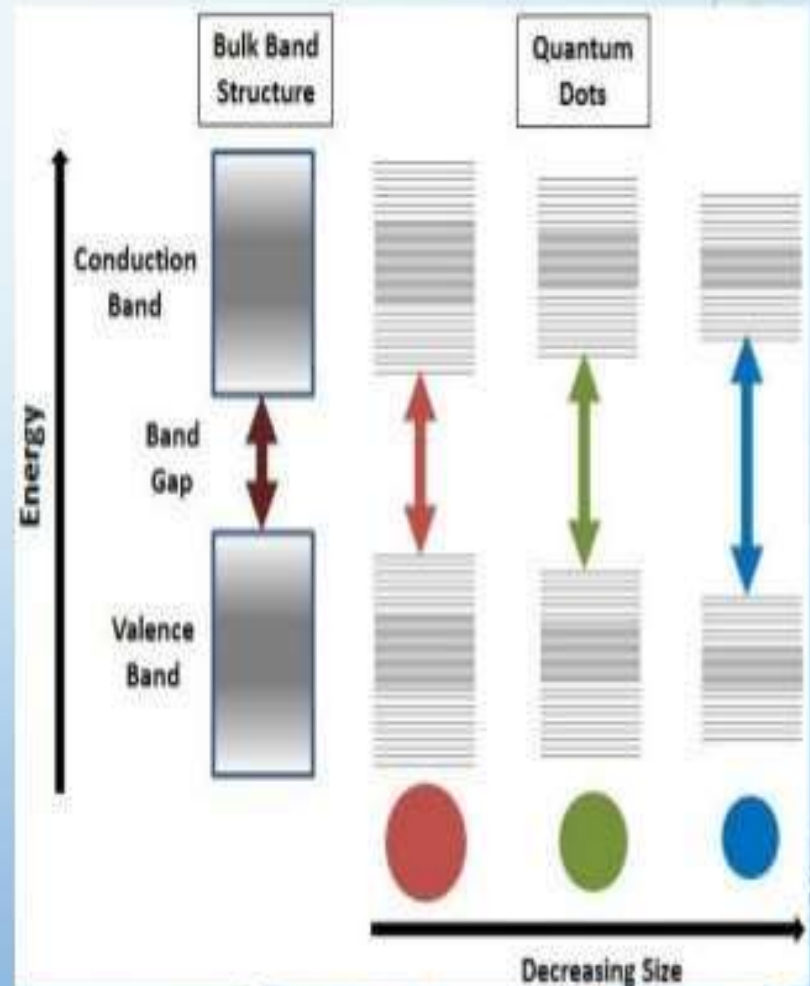
- Nanocrystals
- 2-10 nm diameter
- semiconductors

Effect of quantum confinement: Splitting of energy levels in QD



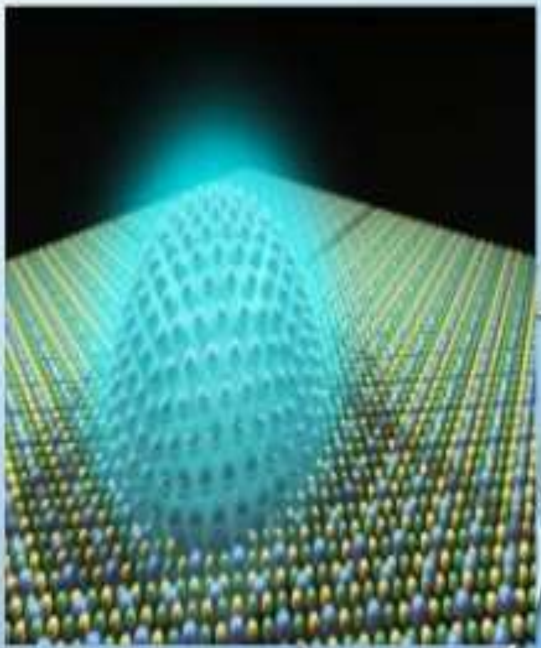
Semiconductor band gap increases with decrease in size of the nanocrystal

- The bandgap of QDs exhibits a never observed before direct relationship to their size.
- The energy band gap is inversely proportional to the size of quantum dots.
- We can vary the energy band gap by changing the size of the quantum dots.
- This property of quantum dots makes it possible to use them in a variety of applications.



WHAT ARE QUANTUM DOTS?

- Quantum dots (QDs) are semiconductor nanocrystals with a core–shell structure and a diameter that typically ranges from 2 to 10 nm.
- They are zero dimensional.
- They display unique electronic properties intermediate between those of bulk semiconductors and discrete molecules.
- They have potential applications in electronic and optoelectronic devices.

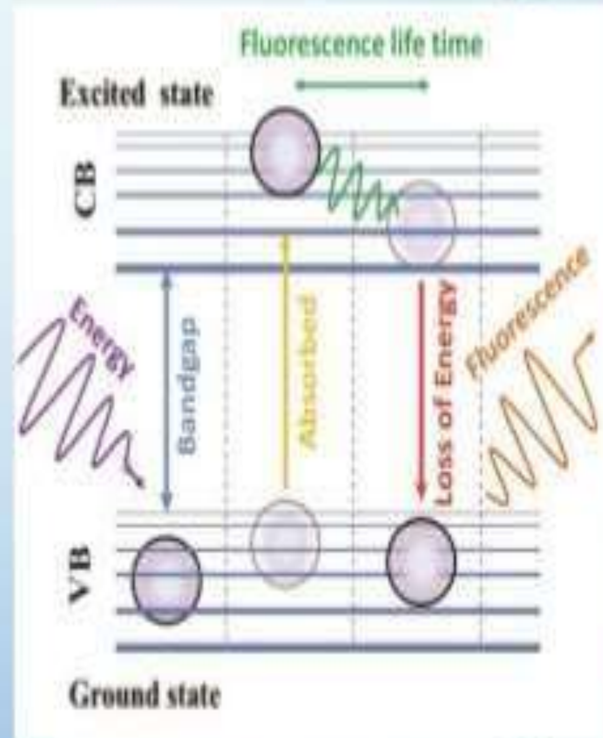


OPTICAL PROPERTIES OF QUANTUM DOTS

- The most characteristic property is that their colour is size dependent and thus can be controlled during synthesis.
- This arises as a result of quantum confinement effect.
- Smaller dots emit higher energy light, that is bluer in colour whereas larger dots emit lower energy red light.

Fig. As size increases a red shift is observed

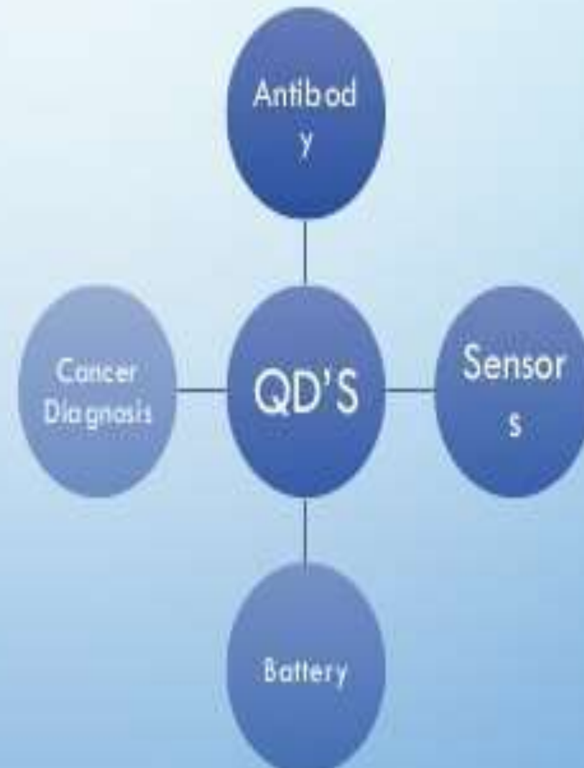
- QD fluorescence occurs when the excited electron moves from the conduction band to its valence band, emitting a photon with a longer wavelength than the one absorbed (electron–hole recombination process).
- QDs can emit light at wavelengths ranging from the ultraviolet (UV) to the infrared (IR).
- The properties of QDs include high photostability, high quantum yield and high molar extinction coefficients (~10–100-times those of organic dyes).
- They also have narrow symmetrical intense emission at specific wavelengths, ranging from the UV to the IR.



APPLICATIONS OF QUANTUM DOTS

Some of the applications of quantum dots include:

- Solar Cells
- Photocatalysts
- Lasers
- QLED
- Computing
- Biology



Potential Uses of Quantum Dots

BioImaging: QDs are coated with hydrophilic polymers and also be conjugated to obtain mulifunctionality. They can be linked with biological agents such as antibodies, peptides, oligonucleotides, inhibitor molecules etc.

E.g. QDs can be used in cancer detection.

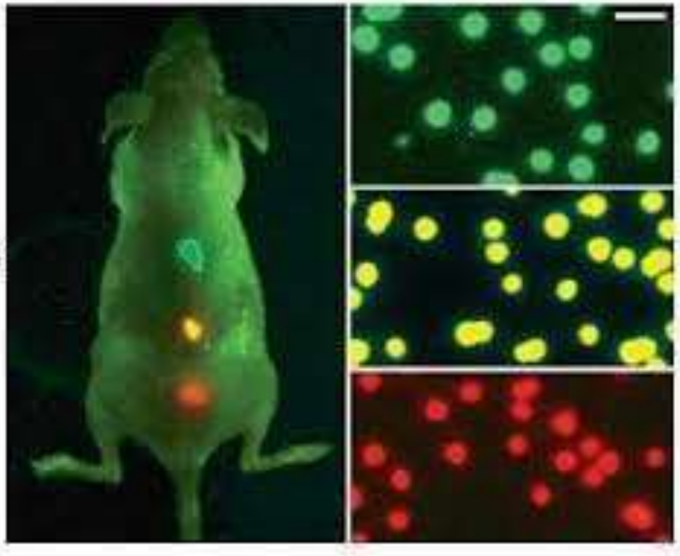
Detection of tumor in human body,

Multicolour Imaging of the tumor tissue by QDs in live animals

In-vivo imaging poses problems with the clearence of the particles from body. Toxicity and hazards effects yet to be evaluated thoroughly

A broad absorption and narrow emission Spectrum means a single excitation source can be used to excite QDs of different colo imaging multiple targets simultaneously.

B



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Potential Uses of Quantum Dots

- **Light Emitting Diodes**

Quantum dot light emitting diodes (QD-LED) and 'QD-White LED' are very useful when producing the displays for electronic devices because they emit light in highly specific Gaussian distributions.

QD-LED displays can render colors very accurately and use much **less power** than traditional displays.
Quantum-dot-based LEDs are characterized by pure and saturated emission colours with narrow bandwidth, and their emission wavelength is easily tuned by changing the size of the quantum dots

- **Solar Cells**

Quantum dot (QD) solar cells have the potential to increase the maximum attainable thermodynamic conversion e fficency of solar photon conversion up to about 66% by utilizing hot photogenerated carriers to produce higher photovoltages or higher photocurrents

Thanks